

Problem Set 3 - Answer sheet

Applied Stats II - Nour Kebbi - 23350337

Due: March 25, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub in .pdf form.
- This problem set is due before 23:59 on Wednesday March 25, 2026. No late assignments will be accepted.

Question 1

We are interested in how governments' management of public resources impacts economic prosperity. Our data come from Alvarez, Cheibub, Limongi, and Przeworski (1996) and is labelled `gdpChange.csv` on GitHub. The dataset covers 135 countries observed between 1950 or the year of independence or the first year for which data on economic growth are available ("entry year"), and 1990 or the last year for which data on economic growth are available ("exit year"). The unit of analysis is a particular country during a particular year, for a total $> 3,500$ observations.

- Response variable:
 - `GDPWdiff`: Difference in GDP between year t and $t-1$. Possible categories include: "positive", "negative", or "no change"
- Explanatory variables:
 - `REG`: 1=Democracy; 0=Non-Democracy
 - `OIL`: 1=if the average ratio of fuel exports to total exports in 1984-86 exceeded 50%; 0= otherwise

Please answer the following questions:

1. Construct and interpret an unordered multinomial logit with `GDPWdiff` as the output and "no change" as the reference category, including the estimated cutoff points and coefficients.

```

1 head(gdp_data)
2 gdp_data$REG <- factor(gdp_data$REG,
3                       levels = c("0", "1"),
4                       labels = c("Non-Democracy",
5                                   "Democracy"))
6 head(gdp_data)
7 gdp_data <- gdp_data %>%
8   mutate(GDPWdiff_category = case_when(
9     GDPWdiff < 0 ~ "negative",
10    GDPWdiff == 0 ~ "no change",
11    GDPWdiff > 0 ~ "positive"))
12 head(gdp_data)
13 gdp_data$GDPWdiff_category <- factor(gdp_data$GDPWdiff_category)
14 gdp_data$GDPWdiff_category <- relevel(gdp_data$GDPWdiff_category, ref = "
    no change")
15 multi.log <- multinom(GDPWdiff_category ~ REG + OIL, data = gdp_data)
16 summary(multi.log)
17 # run model
18 exp(coef(multi.log))
19 # get p values
20 z <- summary(multi.log)$coefficients

```

Interpretation: Holding other factors constant, the odds of a country having positive GDP growth rather than no change are 5.86 times higher for democracy than for non-democracy. Holding other factors constant, the odds of a country having negative GDP growth rather than no change are 3.97 times higher for democracy. The odds of positive GDP growth versus no change is 97.15 times higher for countries exporting oil. The odds of negative GDP growth versus no change is 119.57 times higher for countries exporting oil.

2. Construct and interpret an ordered multinomial logit with `GDPWdiff` as the outcome variable, including the estimated cutoff points and coefficients.

```

1 gdp_data$GDPWdiff_category <- factor(gdp_data$GDPWdiff_category,
2                                     levels = c("negative", "no change",
3                                                 "positive"),
3                                     ordered = TRUE)
4 ord.log <- polr(GDPWdiff_category ~ REG + OIL, data = gdp_data, Hess =
    TRUE)
5 summary(ord.log)
6 exp(coef(ord.log))

```

Interpretation: Holding other variables constant, the odds of a country being in a higher GDP growth are 1.48 times higher for democracrativ countries. The t-value of

democracy is 5.3 hence highly significant. Holding other variables constant, the odds of a country being in a higher GDP growth are 0.8 times lower for countries exporting oil. The t-value of oil is -1.7 so it is not significant.

Question 2

Consider the data set `MexicoMuniData.csv`, which includes municipal-level information from Mexico. The outcome of interest is the number of times the winning PAN presidential candidate in 2006 (`PAN.visits.06`) visited a district leading up to the 2009 federal elections, which is a count. Our main predictor of interest is whether the district was highly contested, or whether it was not (the PAN or their opponents have electoral security) in the previous federal elections during 2000 (`competitive.district`), which is binary (1=close/swing district, 0="safe seat"). We also include `marginality.06` (a measure of poverty) and `PAN.governor.06` (a dummy for whether the state has a PAN-affiliated governor) as additional control variables.

- (a) Run a Poisson regression because the outcome is a count variable. Is there evidence that PAN presidential candidates visit swing districts more? Provide a test statistic and p-value.

```
1 mexico_elections_poisson <- glm(PAN.visits.06 ~ competitive.district +
  marginality.06 + PAN.governor.06,
2                               data = mexico_elections,
3                               family = "poisson")
4 summary(mexico_elections_poisson)
```

Interpretation: The p-value for competitive district is 0.63, hence not very significant.

- (b) Interpret the `marginality.06` and `PAN.governor.06` coefficients.

```
1 exp(coef(mexico_elections_poisson))
```

Interpretation: For every 1 unit increase in the poverty measure, the expected number of visits decreases by 88 percent holding other variables constant. For PAN affiliated governors they receive 27 percent visits less than non-affiliated PAN governors holding other variables constant.

- (c) Provide the estimated mean number of visits from the winning PAN presidential candidate for a hypothetical district that was competitive (`competitive.district=1`), had an average poverty level (`marginality.06 = 0`), and a PAN governor (`PAN.governor.06=1`).

```
1 hypothetical_district <- data.frame(competitive.district = 1,
2                               marginality.06 = 0,
3                               PAN.governor.06 = 1)
4 predict(mexico_elections_poisson, newdata = hypothetical_district, type =
  "response")
```

The estimated mean number of visits from the winning PAN presidential candidate for a hypothetical district that was competitive is 0.01494818.